9/22/2025 | ECC / HW 3 | Walker, Nathan

3.1 Find, by trial and error, a set of four binary codewords of length three such that each word is at least a distance of 2 from every other word.

$[0,0,0]$, $[1,0,1]$, $[1,1,0]$, $[0,1,1]$

3.3 Perhaps the simplest of all codes is the binary parity check code, a $(n, n-1)$ code, where $k = n-1$. Given a message vector $\mathbf{m} = (m_0, m_1, \ldots, m_{k-1})$, the codeword is $\mathbf{c} = (m_0, m_1, \ldots, m_{k-1}, b)$, where $b = \sum_{j=0}^{k-1} m_j$ (arithmetic in $GF(2)$) is the *parity bit*. Such a code is called an even parity code, since all codewords have even parity — an even number of 1 bits.

(a) Determine the minimum distance for this code.
(b) How many errors can this code correct? How many errors can this code detect?
(c) Determine a generator matrix for this code.
(d) Determine a parity check matrix for this code.

(e) Suppose that bit errors occur independently with probability p_e. The probability that a parity check is satisfied is the probability that an even number of bit errors occur in the received codeword. Verify the following expression for this probability:

$$\sum_{i=0,i\text{even}}^{n} \binom{n}{i} p_e^i (1-p_e)^{n-i} = \frac{1+(1-2p_e)^n}{2}.$$

a) 2 b) correct none, can detect 1

c) $mG = C \Rightarrow C = [m_0, m_1, \ldots, m_{k-1} \vdots b]$

$$G = \begin{bmatrix} I_{k-1} & \vdots \\ & 1 \\ & \vdots \\ & 1 \end{bmatrix}_{k \times k+1}$$

d) $H = [I_{n-k} \vdots -P^T] = [I_{n-n+1} \vdots -P^T] = [I_1 \vdots 1 1 \cdots 1]$

$k = n-1$

e) $\sum_{i=0,\text{even}}^{n} \binom{n}{i} p_e^i (1-p_e)^{n-i}$ Compute odd & even sums

even + odd $= (p_e + (1-p_e))^n = 1$ even - odd $= (p_e - (1-p_e))^n = (-1+2p_e)^n$

$$\sum_{i=0,\text{even}}^{n} \binom{n}{i} p_e^i (1-p_e)^{n-i} = \frac{1+(1-2p)^n}{2}$$

3.4 For the $(n, 1)$ repetition code, determine a parity check matrix.

$G = [1 \mid 1\ 1 \ldots 1]_{1 \times n}$ $\quad P = [1\ 1 \ldots 1]_{1 \times n-1}$

$$\left[\begin{array}{c:c} I_{n-1} & -P^T \end{array}\right]$$

i.e. $n=5$ $\left[\begin{array}{cccc:c} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{array}\right]$

3.5 [483] Let $p = 0.1$ be the probability that any bit in a received vector is incorrect. Compute the probability that the received vector contains undetected errors given the following encoding schemes:

(a) No code, word length $n = 8$.
(b) Even parity (see Exercise 3.3), word length $n = 4$.
(c) Odd parity, word length $n = 9$. (Is this a linear code?)
(d) Even parity, word length $= n$.

a) $P(\text{undetected error}) = 1 - P(\text{no bit errors}) = 1 - (1-p)^n = 0.56953279$

b) $\sum_{i=0,\text{even}}^{5} \binom{5}{i} p_c^i (1-p_c)^{5-i} = 0.07335$

c) $\frac{1 + (1-2p)^{10}}{2} - \underbrace{(1-p)^{10}}_{i=0} = 0.205$

d) $\frac{1 + (1-2p)^n}{2} - (1-p)^n$

3.6 [272] Let C_1 be an (n_1, k, d_1) binary linear systematic code with generator $G_1 = [P_1\ I_k]$. Let C_2 be an (n_2, k, d_2) binary linear systematic code with generator $G_2 = [P_2\ I_k]$. Form the parity check matrix for an $(n_1 + n_2, k)$ code as

$$H = \left[\begin{array}{cc} & P_1^T \\ I_{n_1+n_2-k} & I_k \\ & P_2^T \end{array}\right].$$

Show that this code has minimum distance at least $d_1 + d_2$.

$G = [P_1 I_k \ P_2 I_k]$

then new code words are $\underline{C} = [C_1 \mid C_2]$

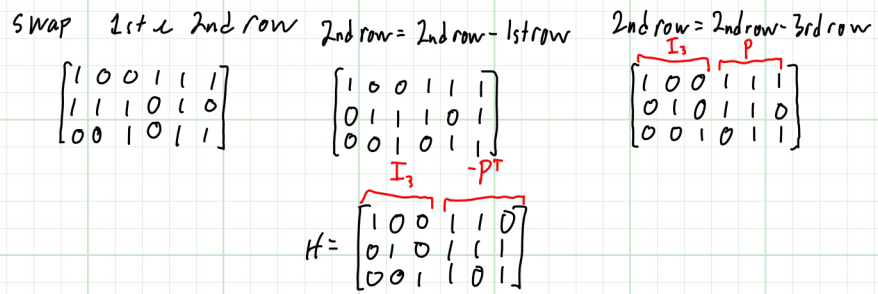
if C_1 has $d_{min} = d_1$ & C_2 has $d_{min} = d_2$

$\therefore$ $\underline{C}$ has $d_{min} = d_1 + d_2$

3.7 The generator matrix for a code over $GF(2)$ is given by

$$G = \begin{bmatrix} 1 & 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}.$$

Find a generator matrix and parity check matrix for an equivalent systematic code.

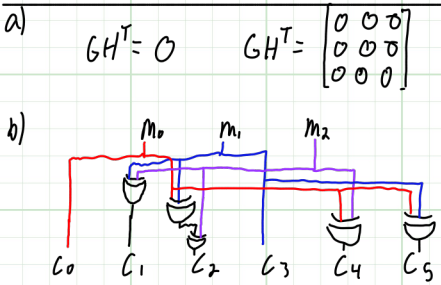


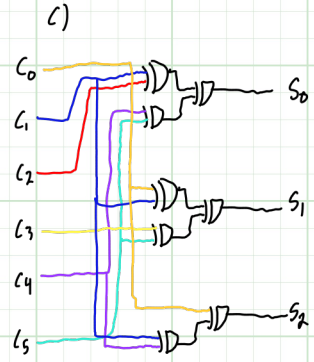
3.8 The generator and parity check matrix for a binary code C are given by

$$G = \begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 1 & 0 \end{bmatrix} \quad H = \begin{bmatrix} 1 & 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 \end{bmatrix}. \tag{3.23}$$

This code is small enough that it can be used to demonstrate several concepts from throughout the chapter.

(a) Verify that H is a parity check matrix for this generator.
(b) Draw a logic diagram schematic for an implementation of an encoder for the nonsystematic generator G using "and" and "xor" gates.
(c) Draw a logic diagram schematic for an implementation of a circuit that computes the syndrome.
(d) List the vectors in the orthogonal complement of the code (that is, the vectors in the dual code $C^{\perp}$).
(e) Form the standard array for code C.
(f) Form the syndrome decoding table for C.
(g) How many codewords are there of weight $0, 1, \ldots, 6$ in C? Determine the weight enumerator $A(z)$.
(h) Using the generator matrix in (3.23), find the codeword with $\mathbf{m} = [1, 1, 0]$ as message bits.
(i) Decode the received word $\mathbf{r} = [1, 1, 1, 0, 0, 1]$ using the generator of (3.23).
(j) Determine the weight enumerator for the dual code.
(k) Write down an explicit expression for $P_u(E)$ for this code. Evaluate this when $p = 0.01$.
(l) Write down an explicit expression for $P_d(E)$ for this code. Evaluate this when $p = 0.01$.
(m) Write down an explicit expression for $P(E)$ for this code. Evaluate this when $p = 0.01$.
(n) Write down an explicit expression for $P(E)$ for this code, assuming a bounded distance decoder is used. Evaluate this when $p = 0.01$.
(o) Write down an explicit expression for $P(F)$ for this code. Evaluate this when $p = 0.01$.
(p) Determine the generator G for an extended code, in systematic form.
(q) Determine the generator for a code which has expurgated all codewords of odd weight. Then express it in systematic form.






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i = 0 dual code word is = [0 0 0 0 0 0]
i = 1 dual code word is = [1 1 0 1 1 0]
i = 2 dual code word is = [1 0 1 0 1 1]
i = 3 dual code word is = [0 1 1 1 0 1]
i = 4 dual code word is = [0 1 0 0 1 1]
i = 5 dual code word is = [1 0 0 1 0 1]
i = 6 dual code word is = [1 1 1 0 0 0]
i = 7 dual code word is = [0 0 1 1 1 0]
i = 0 code word is = [0 0 0 0 0 0]
i = 1 code word is = [1 0 1 0 1 1]
i = 2 code word is = [0 1 1 1 0 1]
i = 3 code word is = [1 1 0 1 1 0]
i = 4 code word is = [0 1 1 0 1 0]
i = 5 code word is = [1 1 0 0 0 1]
i = 6 code word is = [0 0 0 1 1 1]
i = 7 code word is = [1 0 1 1 0 0]
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standard array
    "000000"    "101011"    "011101"    "110110"    "011010"    "110001"    "000111"    "101100"
    "100000"    "001011"    "111101"    "010110"    "111010"    "010001"    "100111"    "001100"
    "010000"    "111011"    "001101"    "100110"    "001010"    "100001"    "010111"    "111100"
    "001000"    "100011"    "010101"    "111110"    "010010"    "111001"    "001111"    "100100"
    "000100"    "101111"    "011001"    "110010"    "011110"    "110101"    "000011"    "101000"
    "000010"    "101001"    "011111"    "110100"    "011000"    "110011"    "000101"    "101110"
    "000001"    "101010"    "011100"    "110111"    "011011"    "110000"    "000110"    "101101"
    "001001"    "100010"    "010100"    "111111"    "010011"    "111000"    "001110"    "100101"
Syndrome Table
    "000000"    "000"
    "100000"    "110"
    "010000"    "101"
    "001000"    "010"
    "000100"    "100"
    "000010"    "111"
    "000001"    "011"
    "001001"    "001"
```

G) wt 0 : 1 , wt 1 : 0 , wt 2 : 0 , wt 3 : 4 , wt 4 : 3 , wt 5 & 6 : 0

$A(z) = 1 + 4z^3 + 3z^4$

H) [0 0 0 1 1 1]

i) [1 1 1 0 0 1] ⇒ Code word = [1 1 0 0 0 1] ⇒ m = [1 0 1]

Used Std Array

j) wt 0 : 1 wt 3 : 4 wt 4 : 3

$A(z) = 1 + 4z^3 + 3z^4$

k) l) m) n) o) Pu = 3.9106e-06 Pd = 0.058516 PE = 0.0013644 bounded PE = 0.0011684 PF = 9.606e-05

P)

add check bit and rearrange

$$\begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 & 1 \end{bmatrix}$$

q) even codewords: [0 0 0 0 0 0] [1 0 1 0 1 1] [0 1 1 1 0 1] [1 1 0 1 1 0] -

$$\begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 0 \end{bmatrix}$$

3.9 [271] Let a binary systematic (8, 4) code have parity check equations

$$c_0 = m_1 + m_2 + m_3$$
$$c_1 = m_0 + m_1 + m_2$$
$$c_2 = m_0 + m_1 + m_3$$
$$c_3 = m_0 + m_2 + m_3.$$

(a) Determine the generator matrix G for this code in systematic form. Also determine the parity check matrix H.
(b) Using Theorem 3.13, show that the minimum distance of this code is 4.
(c) Determine $A(z)$ for this code. Determine $B(z)$.
(d) Show that this is a self-dual code.

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a) I_4 P

$$\begin{matrix} m_0 \\ m_1 \\ m_2 \\ m_3 \end{matrix}\begin{bmatrix} 1&0&0&0&0&1&1&1 \\ 0&1&0&0&1&1&1&0 \\ 0&0&1&0&1&1&0&1 \\ 0&0&0&1&1&0&1&1 \end{bmatrix} = G \qquad H = \begin{bmatrix} 1&0&0&0&0&1&1&1 \\ 0&1&0&0&1&1&1&0 \\ 0&0&1&0&1&1&0&1 \\ 0&0&0&1&1&0&1&1 \end{bmatrix}$$

b) take one column of P since all have w.t 3 results are the same

$$\begin{bmatrix}0\\1\\1\\1\end{bmatrix} = \begin{bmatrix}0\\1\\0\\0\end{bmatrix} + \begin{bmatrix}0\\0\\1\\0\end{bmatrix} + \begin{bmatrix}0\\0\\0\\1\end{bmatrix},$$ can we form w/less? $$\begin{bmatrix}0\\1\\1\\1\end{bmatrix} \neq \begin{bmatrix}1\\1\\1\\0\end{bmatrix} + \begin{bmatrix}1\\1\\0\\1\end{bmatrix} + \begin{bmatrix}1\\0\\1\\1\end{bmatrix} = \begin{bmatrix}3\\2\\2\\2\end{bmatrix} = \begin{bmatrix}1\\0\\0\\0\end{bmatrix}$$

$\therefore d_{min} = 4$

c) $A_4 = 14 \quad A_8 = 1 \quad A_0 = 1$

$$A(z) = B(z) = 1 + 14z^4 + z^8$$

d) Since $G = H$ using H as a generator matrix is the same as using G

3.11 Given a code with a parity check matrix H, show that the coset with syndrome $\mathbf{s}$ contains a vector of weight w if and only if some linear combination of w columns of H equals $\mathbf{s}$.

$\underline{c}H^T = 0$ $\quad$ $\underline{c}$ for some $\underline{z} = \underline{c} + \underline{e}$

$\Downarrow$

$(\underline{c} + \underline{e})H^T = \underline{c}H^T + \underline{e}H^T \Rightarrow 0 + \underline{e}H^T$

wt of $\underline{e} = w$

so $\underline{e}H^T$ is w linear combinations of H

3.12 Show that all of the nonzero codewords of the $(2^m - 1, m)$ simplex code have weight 2^{m-1}. *Hint:* Start with $m = 2$ and work by induction.

$m = 2$ $\quad (3, 2)$

m	code words
00	0 0 0
01	1 0 1
10	0 1 1
11	1 1 0

$G = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$

all code words have $wt\ 2 = 2^{2-1} = 2$

induction $(2^{m+1}-1, m+1)$

$G_{m+1} = \begin{bmatrix} G_m & G_m & \underline{0} \\ \underline{0} & 1 & 1 \end{bmatrix}$

Message - $u = [u_1, u_2, \ldots, u_{m+1}]$ if $u_{m+1} = 0$

then $u_{1-m} \Rightarrow 2^{m-1}$

if $u_{m+1} = 1$ then

$wt = u(u_{1-m}) \cdot 2 = 2^m = 2^{(m+1)-1}$

holds true

3.16 The weight enumerator $A(z)$ of (3.11) for a code C is sometimes written as

$$W_A(x,y) = \sum_{i=0}^{n} A_i x^{n-i} y^i.$$

In this problem, consider binary codes, $q = 2$.

(a) Show that $A(z) = W_A(x,y)|_{x=1,y=z}$.

(b) Let $W_B(x,y) = \sum_{i=0}^{n} B_i x^{n-i} y^i$ be the weight enumerator for the code dual to C. Show that the MacWilliams identity can be written as

$$W_B(x,y) = \frac{1}{q^k} W_A(x+y, x-y)$$

or

$$W_A(x,y) = \frac{1}{q^{n-k}} W_B(x+y, x-y). \tag{3.24}$$

(c) In the following subproblems, assume a binary code. Let $x = 1$ in (3.24). We can write

$$\sum_{i=0}^{n} A_i y^i = \frac{1}{2^{n-k}} \sum_{i=0}^{n} B_i (1+y)^{n-i} (1-y)^i. \tag{3.25}$$

Set $y = 1$ in this and show that $\sum_{i=0}^{n} \frac{A_i}{2^k} = 1$. Justify this result.

(d) Now differentiate (3.25) with respect to y and set $y = 1$ to show that

$$\sum_{i=1}^{n} \frac{iA_i}{2^k} = \frac{1}{2}(n - B_1).$$

If $B_1 = 0$, this gives the average weight.

(e) Differentiate (3.25) v times with respect to y and set $y = 1$ to show that

$$\sum_{i=v}^{n} \binom{i}{v} \frac{A_i}{2^k} = \frac{1}{2^v} \sum_{i=0}^{v} (-1)^i \binom{n-i}{v-i} B_i.$$

Hint: Define $(x)_+^n = \begin{cases} 0 & n < 0 \\ x^n & n \geq 0 \end{cases}$. We have the following generalization of the product rule for differentiation:

$$\frac{d^v}{dy^v}(y+a)^q (y+b)^q = \sum_{j=0}^{v} \binom{v}{j} \frac{p!}{(p-j)!} (y+a)_+^{p-j} \frac{q!}{(q-(v-j))!} (y+b)_+^{q-(v-j)}.$$

(f) Now set $y = 1$ in (3.24) and write

$$\sum_{i=0}^{n} A_i x^{n-i} = \frac{1}{2^{n-k}} \sum_{i=0}^{n} B_i (x+1)^{n-i} (x-1)^i.$$

Differentiate v times with respect to x and set $x = 1$ to show that

$$\sum_{i=0}^{k-v} \binom{n-i}{v} A_i = 2^{k-v} \sum_{i=0}^{v} \binom{n-i}{n-v} B_i, \quad v = 0, 1, \ldots, n. \tag{3.26}$$

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a) $W_A(1, z) = \sum_{i=0}^{n} A_i (1)^{n-i} z^i = \sum_{i=0}^{n} A_i z^i$ — z^i of wt i = coeff of wt i $= A(z)$

b) $(x+y) + (x-y) = (1 + (q-1)z) + (1-z)$ $\qquad (x+y) - (x-y) = (1 + (q-1)z) - (1-z)$

$2x = 2((q-1)z - z)$ $\qquad 2y = qz$

$x = 1 + \frac{q-2}{2} z$ $\qquad y = \frac{q}{2} z$

$$W_A\left(1 + \tfrac{q-2}{2} z, \tfrac{q}{2} z\right) = \sum_{i=0}^{n} A_i \left(1 + \tfrac{q-2}{2} z + \tfrac{q}{2} z\right)^{n-i} \left(1 + \tfrac{q-2}{2} z - \tfrac{q}{2} z\right)^i$$

$$\sum_{i=0}^{n} A_i \left(1 + (q-1)z\right)^{n-i} (1+z)^i$$

cubb q^{-k} and W_A leads to $B(z)$

$$(1 + (q-1)z)^n \sum_{i=0}^{n} A_i \left(\frac{1+(q-1)z}{1+z}\right)^i (1+z)^i$$

same process for $A(z)$

c) $\frac{1}{2^{n-k}} \sum_{i=0}^{n} B_i (1+y)^{n-i} (1-y)^i = \sum_{i=0}^{n} A_i y^i$ $\qquad y = 1$

$\frac{1}{2^{n-k}} \sum_{i=0}^{n} B_i (2)^{n-i} (z)^i = \sum_{i=0}^{n} A_i$

$2^k \sum_{i=0}^{n} B_i = \sum_{i=0}^{n} A_i \Rightarrow \sum_{i=0}^{n} \frac{A_i}{2^k} \Rightarrow \sum_{i=0}^{n} A_i = |C| = 2^k$

$\frac{1}{2^k} \sum_{i=0}^{n} A_i = 1$

d) $\frac{\partial}{\partial y} \frac{1}{2^{n-k}} \sum_{i=0}^{n} B_i (1+y)^{n-i} (1-y)^i = \frac{1}{2^{n-k}} \sum_{i=0}^{n} B_i \left[(1+y)^{n-i-1} (1-y)^{i-1} [(n-2i) - ny]\right]$ $\qquad y = 1$

derivative only exists for $i = 0$ & 1 so redifferentiate

$\frac{1}{2^{n-k}} \left(B_0 n 2^{n-1} - B_1 2^{n-1}\right) \Rightarrow$ $\qquad (n - B_1)\tfrac{1}{2} = \sum_{i=0}^{n} \frac{iA_i}{2^k}$ ✓ trivial differentiation

e)

$$\frac{1}{dy^v}\left[\sum_{i=0}^{n} A_i y^i = \frac{1}{2^{n-k}} \sum_{i=0}^{n} B_i (1+y)^{n-i}(1-y)^i\right]$$

LHS

$$\sum_{i=v}^{n} A_i \frac{i!}{(i-v)!} y^{i-v} = \sum_{i=v}^{n} \binom{i}{v} A_i y^{i-v}$$

RHS

$$\frac{1}{2^{n-k}} \sum_{i=0}^{v} B_i \binom{v}{i} (n-i)_i (-1)^i i! \, 2^{n-v}$$

$$v! \, 2^{k-v} \sum_{i=0}^{v} (-1)^i \binom{n-i}{v-i} B_i$$

$$\frac{1}{2^k} \sum_{i=v}^{n} \binom{i}{v} A_i y^{i-v} = \frac{1}{2^v} \sum_{i=0}^{v} (-1)^i \binom{n-i}{v-j} B_i$$

3.18 [272] Let C be a binary linear code with both even- and odd-weight codewords. Show that the number of even-weight codewords is equal to the number of odd-weight codewords.

$E = \{$even-weight codewords of $C\}$

$O = \{$odd-weight codewords of $C\}$

show $|E| = |O|$

$x \in O \qquad f: C \to C, \; f(c) = c + x$

$f(c_1) = f(c_2) : c_1 + x = c_2 + x$

$c_1 = c_2$

$y \in C$, take $c = y + x \in C$

$f(c) = c + x = y + x + x = y$ since in $GF(2)$

if $c \in E$ then

$wt(f(c)) = wt(c + x)$

$c \in E \Rightarrow f(c) \in O, \quad c \in O \Rightarrow f(c) \in E$

3.19 Show that for a binary code, $P_u(E)$ can be written as:

(a) $P_u(E) = (1-p)^n \left[A\left(\frac{p}{1-p}\right) - 1\right]$

(b) and $P_u(E) = 2^{k-n} B(1-2p) - (1-p)^n$.

$$P_U(E) = \sum_{j=d_{min}}^{n} A_j p^j (1-p)^{n-j} = (1-p)^n \sum_{j=d_{min}}^{n} A_j p^j (1-p)^{-j}$$

(#wb of j)

$$A\left(\frac{p}{1-p}\right) = 1 + A_1 \left(\frac{p}{1-p}\right)^1 \ldots + A_n \left(\frac{p}{1-p}\right)^n$$

The sum just computes $A\left(\frac{p}{1-p}\right)$ at a particular value if we subtract the 0wt c.w. then they are equivalent

$$\therefore (1-p)^n \sum_{j=d_{min}}^{n} A_j p^j (1-p)^{-j} = (1-p)^n \left[A\left(\frac{p}{1-p}\right) - 1\right]$$

b) $$P_U(E) = \sum_{i=1}^{n} B_i p^i (1-p)^{n-i} =$$

using macWilliams identity

$$\sum_{i=1}^{n} B_i p^i (1-p)^{n-i} = 2^{k-n} \sum_{i=0}^{n} B_i (1-2p)^i - (1-p)^n$$

3.26 [272] Let Γ be the ensemble of *all* the binary systematic linear (n, k) codes.

(a) Prove that a nonzero binary vector $\mathbf{v}$ is contained in exactly $2^{(k-1)(n-k)}$ of the codes in Γ or it is in none of the codes in Γ.

(b) Using the fact that the nonzero n-tuples of weight $d-1$ or less can be in at most

$$2^{(k-1)(n-k)} \sum_{i=1}^{d-1} \binom{n}{i}$$

(n, k) systematic binary linear codes, show that there exists an (n, k) linear code with minimum distance of at least d if the following bound is satisfied:

$$\sum_{i=1}^{d-1} \binom{n}{i} < 2^{n-k}.$$

(c) Show that there exists an (n, k) binary linear code with minimum distance at least d that satisfies the following inequality:

$$2^{n-k} \leq \sum_{i=0}^{d} \binom{n}{i}.$$

This provides a lower bound on the minimum distance attainable with an (n, k) linear code known as the **Gilbert–Varshamov bound**.

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a)

$|\Gamma| = 2^{k(n-k)}$ all possible matrices to reside in Γ

$\underline{V} \neq \underline{0}$ $\quad \underline{V} = \underline{m}\Gamma$ $\quad \underline{m}$ · all possibilities

for systematic $[V_1 : V_2] = [\underline{m} ; \underline{m}P]$ $\quad P_{k \times (n-k)}$

$V_2 = mP$ Sum of $wt(m) \geq 1$ rows must equal V_2 there are at least $k-1$ rows not used if $wt(m) = 1$

$\therefore$ there are $2^{(k-1)(n-k)}$ possibilities (rows: length of rows)

V is in Γ if not then it is in none of them

b)

$2^{k(n-k)} = 2^{(k-1)(n-k)} \cdot 2^{n-k}$ (total codewords)

$2^{(k-1)(n-k)} \sum_{i=1}^{d-1} \binom{n}{i} \leq 2^{(k-1)(n-k)} \cdot 2^{n-k}$

$\sum_{i=1}^{d-1} \binom{n}{i} \leq 2^{n-k}$

c) $C \subseteq \{0,1\}^n$ $\quad \dim = k$ $\quad |C| = 2^k$ $\quad d_{min} \geq d$

$2^{n-k} \leq \sum_{i=0}^{d} \binom{n}{i}$

$V_2(n, d-1) = \sum_{i=0}^{d-1} \binom{n}{i} \cdot 2^k \leq 2^n$

$\sum_{i=0}^{d-1} \binom{n}{i} \leq 2^{n-k}$

Use $V_2(n, d) \geq V_2(n, d-1)$

$\therefore \sum_{i=0}^{d} \binom{n}{i} \geq 2^{n-k}$

3.27 Define a linear (5, 3) code over $GF(4)$ by the generator matrix

$$G = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 2 \\ 0 & 0 & 1 & 1 & 3 \end{bmatrix}.$$

(a) Find the parity check matrix.
(b) Prove that this is a single-error-correcting code.
(c) Prove that it is a double-erasure-correcting code.
(d) Prove that it is a perfect code.

---

a)

$$P = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \qquad H = \left[\begin{array}{cc|ccc} 1 & 0 & 3 & 3 & 3 \\ 0 & 1 & 3 & 2 & 1 \end{array}\right]$$

b) # of dependent columns = 3 = dmin

can correct $\left\lfloor \frac{d_{min}-1}{2} \right\rfloor = \left\lfloor \frac{3-1}{2} \right\rfloor = 1$

c) We need 2 columns of G to be lin. indep.

$G = [\underline{g}_1\ \underline{g}_2\ \underline{g}_3\ \underline{g}_4\ \underline{g}_5]$ $\quad g_1, g_2, g_3$ are lin. indep.

$a\underline{g}_i + b\underline{g}_4 = 0$ iff $a = b = 0$ $\quad i = 1,2,3$

$a\underline{g}_i + b\underline{g}_5 = 0$ iff $a = b = 0$ $\quad i = 1,2,3$

$a\underline{g}_4 + b\underline{g}_5 = 0$ iff $a = b = 0$

All pairs of columns are lin. ind.

∴ it is a double erasure code

d) $q = 4 \quad k = 3 \quad n = 5 \quad d = 3 \quad t = 1$

$$q^k \cdot \sum_{i=0}^{t} \binom{n}{i} (q-1)^i \leq q^n$$

$64 \cdot 16 = 1024 \leq (4)^5 = 1024$

yes it is perfect

work through proof of Lemma 3.29

Lemma 3.29 *If C is an (n, k) binary linear code and f is a function defined on $\mathbb{F}_2^n$,*

$$\sum_{\mathbf{u}\in C^\perp} f(\mathbf{u}) = \frac{1}{|C|}\sum_{\mathbf{u}\in C}\hat{f}(\mathbf{u}) \quad \textit{or} \quad \sum_{\mathbf{u}\in C} f(\mathbf{u}) = \frac{1}{|C^\perp|}\sum_{\mathbf{u}\in C^\perp}\hat{f}(\mathbf{u}).$$

Here $|C|$ denotes the number of elements in the code C.

Proof *of lemma*.

$$\sum_{\mathbf{u}\in C}\hat{f}(\mathbf{u}) = \sum_{\mathbf{u}\in C}\sum_{\mathbf{v}\in\mathbb{F}_2^n}(-1)^{\langle\mathbf{u},\mathbf{v}\rangle}f(\mathbf{v}) = \sum_{\mathbf{v}\in\mathbb{F}_2^n} f(\mathbf{v})\sum_{\mathbf{u}\in C}(-1)^{\langle\mathbf{u},\mathbf{v}\rangle}$$
$$= \sum_{\mathbf{v}\in C^\perp} f(\mathbf{v})\sum_{\mathbf{u}\in C}(-1)^{\langle\mathbf{u},\mathbf{v}\rangle} + \sum_{\mathbf{v}\in C\setminus\{\mathbf{0}\}} f(\mathbf{v})\sum_{\mathbf{u}\in C}(-1)^{\langle\mathbf{u},\mathbf{v}\rangle},$$

where $\mathbb{F}_2^n$ has been partitioned into two disjoint sets, the dual code $C^\perp$ and the nonzero elements of the code, $C\setminus\{\mathbf{0}\}$. In the first sum, $\langle\mathbf{u},\mathbf{v}\rangle = 0$, so the inner sum is $|C|$. In the second sum, for every $\mathbf{v}$ in the outer sum, $\langle\mathbf{u},\mathbf{v}\rangle$ takes on the values 0 and 1 equally often as $\mathbf{u}$ varies over C in the inner sum, so the inner sum is 0. Therefore,

$$\sum_{\mathbf{u}\in C}\hat{f}(\mathbf{u}) = \sum_{\mathbf{v}\in C^\perp} f(\mathbf{v})\sum_{\mathbf{u}\in C}(1) = |C|\sum_{\mathbf{v}\in C^\perp} f(\mathbf{v}).$$

□

work through thm 3.28

Proof of Theorem 3.28. The weight enumerators of the code and its dual can be written as

$$A(z) = \sum_{\mathbf{u}\in C} z^{\mathrm{wt}(\mathbf{u})} \quad B(z) = \sum_{\mathbf{u}\in C^\perp} z^{\mathrm{wt}(\mathbf{u})}.$$

Let $f(\mathbf{u}) = z^{\mathrm{wt}(\mathbf{u})}$. Its Hadamard transform is

$$\hat{f}(\mathbf{u}) = \sum_{\mathbf{v}\in\mathbb{F}_2^n}(-1)^{\langle\mathbf{u},\mathbf{v}\rangle}z^{\mathrm{wt}(\mathbf{v})}.$$

Writing out the inner product and the weight function explicitly on the vectors $\mathbf{u} = (u_0, u_1, \ldots, u_{n-1})$ and $\mathbf{v} = (v_0, v_1, \ldots, v_{n-1})$, we have

$$\hat{f}(\mathbf{u}) = \sum_{\mathbf{v}\in\mathbb{F}_2^n}(-1)^{\sum_{i=0}^{n-1}u_iv_i}\prod_{i=0}^{n-1}z^{v_i} = \sum_{\mathbf{v}\in\mathbb{F}_2^n}\prod_{i=0}^{n-1}(-1)^{u_iv_i}z^{v_i}.$$

The sum over the 2^n values of $\mathbf{v}\in\mathbb{F}_2^n$ can be expressed as n nested summations over the binary values of the elements of $\mathbf{v}$:

$$\hat{f}(\mathbf{u}) = \sum_{v_0=0}^{1}\sum_{v_1=0}^{1}\cdots\sum_{v_{n-1}=0}^{1}\prod_{i=0}^{n-1}(-1)^{u_iv_i}z^{v_i}.$$

Now the distributive law can be used to pull the product out of the summations,

$$\hat{f}(\mathbf{u}) = \prod_{i=0}^{n-1}\sum_{v_i=0}^{1}(-1)^{u_iv_i}z^{v_i}.$$

If $u_i = 0$, then the inner sum is $1 + z$. If $u_i = 1$, then the inner sum is $1 - z$. Thus,

$$\hat{f}(\mathbf{u}) = (1+z)^{n-\mathrm{wt}(\mathbf{u})}(1-z)^{\mathrm{wt}(\mathbf{u})} = \left(\frac{1-z}{1+z}\right)^{\mathrm{wt}(\mathbf{u})}(1+z)^n. \tag{3.14}$$

The weight enumerator for the dual code is

$$B(z) = \sum_{\mathbf{u}\in C^\perp} z^{\mathrm{wt}(\mathbf{u})} = \sum_{\mathbf{u}\in C^\perp} f(\mathbf{u}) = \frac{1}{|C|}\sum_{\mathbf{u}\in C}\hat{f}(\mathbf{u}) \quad \text{(by Lemma 3.29)}$$
$$= \frac{1}{|C|}\sum_{\mathbf{u}\in C}\left(\frac{1-z}{1+z}\right)^{\mathrm{wt}(\mathbf{u})}(1+z)^n \quad \text{(by (3.14))}$$
$$= \frac{1}{|C|}(1+z)^n\sum_{\mathbf{u}\in C}\left(\frac{1-z}{1+z}\right)^{\mathrm{wt}(\mathbf{u})} = \frac{1}{|C|}(1+z)^nA(z)\Big|_{z=(1-z)/(1+z)}.$$

□